

Payoff scale reshapes how language models play the prisoner’s dilemma

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Abstract

Large language models are increasingly deployed as agents that negotiate and decide for their users, so whether they cooperate is a governance question. We ask how two everyday levers shape it: the size of what is at stake, and the language the game is posed in. The first is unusual: multiplying every payoff by a positive constant leaves preferences, best replies, equilibria and the orbits of the replicator dynamics untouched, so its predicted behavioural effect is zero by theorem, not convention. Using it as an instrument, five frontier models played an iterated prisoner’s dilemma across ten payoff scales, crossed with prompt language and persona, on a balanced grid of 200,000 decisions. Cooperation moves in every model, by up to 0.636 (95% CI 0.588, 0.686), differently in each, so averaging over models reports a movement no model makes. Its size depends on the prompt language; the scale also gates the persona instruction, reversing its sign in two models of five where every printed payoff is at most 1. The effect is already present on the opening move, and what is played changes too. Cooperation rates published for language models should be read, and governed, alongside their sensitivity to payoff scale.

Subject areas: artificial intelligence, game theory, multi-agent systems

Keywords: machine behaviour, large language models, iterated prisoner’s dilemma, payoff scaling, utility invariance, cross-lingual behaviour

1 Introduction

Large language models are increasingly deployed as *agents* that negotiate, coordinate and decide on behalf of their users, in recommendation systems, negotiation tools and multi-agent assistants [1–4]. In these settings behaviour emerges from strategic interaction rather than

from isolated model outputs, and the agents meet *cooperation dilemmas* whose resolution bears directly on safety, coordination and AI governance [5–8]. A fast-growing literature measures how they behave when they do, with the instruments of behavioural economics and cognitive psychology [9–11] and with the classical games [12]. Equally consequential is how far that behaviour adapts to the costs and benefits at issue, since a deployed agent meets the same dilemma at many different magnitudes [3].

A growing body of work evaluates language models through game-theoretic lenses, and in particular through matrix and repeated games, reporting systematic departures from equilibrium play, persistent cooperative biases and sensitivity to contextual factors such as language and incentives [13–20]. The prisoner’s dilemma is the workhorse of that literature, its strategy space mapped and its predictions sharp [21–26]. Within it the models turn out to be individual rather than uniform, differing in how readily they retaliate [27], how readily they forgive [28], how much weight they give the game rather than the story around it [29] and how far an instructed disposition carries into conditioned reciprocity [30]. These studies share two unremarked reporting conventions. Each summarizes a model by a cooperation rate, or a few of them, measured at one payoff matrix; and each reports what the agents did rather than the decision rule that produced it.

Evaluating strategic behaviour at the level that alignment and governance require therefore asks for more than aggregate outcomes. Drawing on behavioural and evolutionary game theory [31], a behavioural intention can be conceptualised as an agent’s *strategy*: a decision rule mapping an interaction history to the next action [32–35]. The canonical strategies of repeated games, Always Cooperate, Always Defect, Tit-for-Tat and Win-Stay-Lose-Shift [31, 36], supply an interpretable vocabulary for such intentions, and prior work has shown that they can be inferred from noisy behavioural trajectories by supervised learning [32, 33] and extended by probabilistic models to mixed or stochastic play [37]. The inference is not free. Separating the memory-one rules from observed play has been shown to need a long run of observations even where the resident rule is fixed by construction [38], which is why the read-out used here is reported as a resemblance, with its provenance beside it (§2.3).

In parallel, what a language model does is not invariant across the language it is addressed in [13]. Strategic reasoning, risk sensitivity and cooperation patterns depend on linguistic framing, sometimes by margins comparable to the differences between models [29, 39, 40]. Whether the strategies agents play are themselves language-dependent, and whether that dependence introduces systematic bias into multi-agent settings, is largely open.

That dependence is one instance of a wider pattern: what is measured on a language model depends on things the measurement was not about. Accuracy moves with the order of the in-context examples [41], and with formatting choices that leave the meaning of the prompt intact, by margins wide enough to swamp the differences between the models being compared [42]. Instructions about who the model is are equally consequential: a persona changes what a model produces [43], shifts measured capability [44] and, conditioned on a demographic backstory, reproduces the response distribution of the matching human subgroup [45]. What none of these manipulations can supply is a null.

The choice of units is not usually stated as a design decision at all, because in the theory it is not one. A payoff matrix is a von Neumann–Morgenstern utility, defined only up to a positive affine transformation [46, 47], so writing the same dilemma with cells ten times larger changes no preference, no best reply, no equilibrium and no orbit of the replicator dynamics. Under the

theory the two matrices are not two games written differently. They are one game. The theory of the iterated dilemma inherits the indifference, turning on the ordering of the four payoffs and on ratios among them and never on their absolute size [21, 23, 24, 26], and the modelling literature normalizes the scale away [48, 49]. Current theory works at the indifference rather than merely permitting it: where social learners update on the payoff they last received, the condition for conditional cooperation to persist under strong selection loses the benefit and the cost of cooperation altogether and turns on the ordering of the payoffs alone [50].

The invariance belongs to the theory, not to people: human valuation is neither linear in magnitude nor anchored at zero [51], and human choice reverses between logically equivalent descriptions of one problem [52]. That gap has been argued to be structural rather than incidental, with the linguistic description of an action entering the utility function directly and a shift proposed from outcome-based to language-based preferences [53].

Two things are established, and they have not been put together. The first is that what a language model does depends on features of the prompt that carry no task content: the order of the examples, the formatting, the persona, the language. The second is that multiplying every payoff by a positive constant carries no task content in the strictest sense available, because the matrix it produces is not a second game but the same one. What is not known is whether the strategic behaviour of a language model respects that invariance. To our knowledge, no study has asked. Every manipulation in the sensitivity literature could legitimately change behaviour, so a measured change under any of them is at worst a nuisance and at best a finding, against an expected effect nobody had specified. The same holds of the human framing results that motivate language-based preferences, where a label such as donate rather than boost carries connotation, so what is under test there is an assumption about preferences and not a theorem about their representation [53]. A rescaling is the one manipulation whose expected effect is specified, and specified as zero, so a measured departure has a stated reference value rather than a comparison condition.

What follows if the invariance fails is a question about what a published cooperation rate means. Machine behaviour proposes that autonomous systems be studied empirically, with the instruments of the behavioural sciences, because what such a system does is not deducible from how it was built [54]. Those instruments arrive with a known difficulty: a behavioural manipulation almost never carries a predicted effect, so a measured change has to be read against a comparison condition the experimenter chose. The invariance classes of decision theory are the exception, supplying a manipulation whose predicted effect is a stated number, and the number is zero.

The consequences run through the strands of machine behaviour this study touches. The persona line is the one place where the experimenter addresses the agent about its social role, and whether it takes hold turns out to depend on the units the matrix was printed in. The departure is already present on the opening move, which locates it in the reading of the matrix rather than in the weighing of a history. We report no evolutionary dynamics of our own; what we report is a constraint on an input such dynamics take, since models of mixed human-machine populations carry the cooperativeness of the artificial agent as a parameter and ask where the population settles [55–59]. A rate partly determined by the experimenter’s units is not the quantity those models need, and if the dependence takes a different form in each model it is invisible to exactly the pooled designs the field uses to summarize many models at once.

Together, these considerations let us put four questions to the rescaling:

- (1) Does strategic behaviour survive multiplying every payoff by a positive constant, and by how much does it fail to?
- (2) Does the answer depend on the prompt language, and on the persona the model is told to adopt?
- (3) Does the rescaling change only how often a model cooperates, or also the strategy it is playing?
- (4) Does the effect enter at the opening move, or through the course of play?

We answer them on a complete and exactly balanced grid of 10,000 dyads (§2).

Here, we turn a theorem into an instrument. The contributions of the present paper are: **(i)** a payoff-rescaling protocol whose predicted behavioural effect is exactly zero, used to ask whether five frontier language models are playing the game they were shown (§2.2); all five depart from the invariance, by up to 0.636 in cooperation. **(ii)** a per-model analysis showing that the departure has a different shape in each model, to the point of anticorrelation, so that what a pooled analysis reports is a movement no model in the pool makes: the interaction of the payoff scale with the model carries 17.9% of the variance across cells against 5.8% for the main effect, and averaging the five curves gives a range of 0.154 where the mean within-model range is 0.249. The main effect is significant here only because one model departs far enough to survive the averaging, which is a fact about that model rather than a property of language models in general, and a panel assembled without it returns no significant main effect at all. The same holds one level down, since the size of the departure depends on the prompt language as well as on the model, so that per-model, per-language reporting is a matter of validity rather than of presentation. **(iii)** evidence that the departure is structural and not a shift in level: the payoff scale gates the persona instruction, reversing its sign in two models of five and abolishing it in a third, play becomes measurably less consistent with any canonical memory-one rule in four models of five while the fifth shows no trend, and the effect is already present on the opening move, before any history exists.

An earlier and smaller collection on this project has been reported separately by the present authors [60]: three models at three payoff scales, in the same five prompt languages, with cooperation read off the trajectories and reported as rising with the stakes. The present article differs in what it collects and in what it claims. It collects six models rather than three and ten payoff scales rather than three, spanning five orders of magnitude, on a grid whose completeness and balance are enforced by the analysis build rather than assumed. It frames the rescaling as an invariance whose predicted effect is a theorem rather than as a manipulation of the stakes whose effect was expected. And it reports every quantity per model, which is what shows the movement to be heterogeneous, with one of the five running downwards, and the pooled statement to be an artefact of averaging.

2 Material and methods

2.1 Framework

This study runs the protocol of FAIRGAME, the Framework for AI Agents Bias Recognition using Game Theory [13], which supports systematic and reproducible evaluation of language

models through controlled game-theoretic experiments: an experimental condition fixes the payoff structure, the horizon, the model backend and the prompt language, and the repeated normal-form game is then simulated and logged round by round. We did not run the reference implementation. The models studied here are served through a benchmarking platform the reference implementation does not address, so the protocol was reimplemented against that platform, reproducing FAIRGAME’s language-specific prompt templates byte for byte, its reply parsing and its penalty scoring, and writing the same per-game table schema. The templates are compared against the reference files before every collection run, because one save in the wrong encoding corrupts the prompt that reaches the model rather than a comment beside it. Three things are ours rather than the protocol’s. First, the payoff-scaling factor applied to the prisoner’s dilemma, which is the manipulation this paper is about (§2.2). Second, a completeness guard that refuses to release an analysis table unless the grid is exactly balanced and internally consistent. Third, a strategy read-out that reports which canonical memory-one rule a trajectory most resembles, and with what provenance (§2.3).

The design is a complete factorial crossing of six language models, ten payoff scales, five prompt languages, four persona pairings and ten replicates.

The unit of collection is the dyad, one ten-round game between two agents of the same model, contributing two agent-games, an agent-game being the ten decisions one agent takes in one game. Each of the 300 model-by-language-by-scale cells holds exactly 40 dyads, the four persona pairings by ten replicates; pooling the five languages gives 200 dyads and 400 agent-games in each of the 60 model-by-scale cells. The corpus is 12,000 dyads, 24,000 agent-games and 240,000 decisions.

Five of the six models are reported in the main text and the sixth, Gemini 3.1 Flash-Lite, in the electronic supplementary material, for the reason given with the models below. The five carry 250 of the cells, 10,000 dyads, 20,000 agent-games and 200,000 decisions, and are the grid every main-text figure and table is computed on. Balance is enforced rather than assumed: the build refuses to write its tables unless every cell is complete and each game’s recorded payoffs recover the scale its cell was run at (electronic supplementary material, §S1).

2.2 The payoff-scaled prisoner’s dilemma

Both agents receive the prisoner’s-dilemma prompt of the FAIRGAME protocol [13] in its no-communication, announced-horizon condition. It casts the game as a criminal interrogation, states the four payoffs as penalties, names the two actions OptionA and OptionB, carries the running history of play, announces the ten-round horizon and instructs the agent to minimize its penalty.

At $\lambda = 1$ the agent is shown $T = 0$, $R = 2$, $P = 6$, $S = 10$ in penalty units, in the usual temptation, reward, punishment and sucker notation. The ordering $T < R < P < S$ is the image of the familiar $T > R > P > S$ on utilities $u = -\text{penalty}$, and $2R > T + S$ holds on those utilities, so alternation is not preferred to mutual cooperation. OptionA pays less in both columns and is therefore the dominant, defecting action, while mutual OptionB at 2 beats mutual OptionA at 6 and is the cooperative one; cooperation means OptionB throughout this paper. The polarity was checked against the data as well as against the prompt [60] (electronic supplementary material, §S1).

The manipulation multiplies all four cells by a positive constant λ (figure 1a), taking ten

values spanning five orders of magnitude, $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$, and changes nothing else in the prompt (electronic supplementary material, §S2). Two details bear on interpretation: $T = 0$ prints as 0 at every scale, so only three of the four numbers move, and the history block records realised penalties, so λ is present in every prompt of the game and not only the first. We treat the printed penalties as the agent’s utilities, as the prompt’s minimization instruction directs; because λu with $\lambda > 0$ is a positive affine transformation of a von Neumann–Morgenstern utility [46, 47], the predicted effect on an agent playing the game is exactly zero. It is this that separates the present manipulation from a manipulation of the stakes, whose behavioural effect is expected rather than specified [51, 52]. An agent that instead applies a non-linear valuation to the printed penalties is not bound by the invariance; §4 returns to readings of that kind.

Six frontier models were run, each against itself; two different models were never paired. They were Claude Haiku 4.5, GPT-5.4 Nano, Gemini 3.5 Flash-Lite, Qwen3 235B-A22B, Grok 4.20 and Gemini 3.1 Flash-Lite, served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint. Their endpoint identifiers, four dated and two not, are in the electronic supplementary material, §S2.

The main text reports the first five of them. The sixth, Gemini 3.1 Flash-Lite, is reported in full in the electronic supplementary material, §S8, and the reason is a property of its endpoint rather than of its results: its identifier names an endpoint that is both undated and announced as provisional, so what was served under that name while the corpus was collected cannot be re-addressed afterwards. We state the cost of drawing the line there rather than leaving it to be found. `gemini-3.5-flash-lite`, which stays in the main text, is an undated alias as well and could in principle be re-pointed to a later build, so its results carry a weaker form of the same risk; the line drawn here is between an identifier marked provisional and one that is not, and not between pinned and unpinned. It is also the one model of the six whose strategy trend runs opposite to the rest (§3), which is a reason to report it prominently rather than to set it aside. Every quantity the main text gives for the five is given for all six in the electronic supplementary material, which also states what including it changes, and on every conclusion of this paper that is nothing.

The prompt languages are English, French, Vietnamese, Chinese and Arabic, in templates byte-identical to the FAIRGAME resource files and checked against them before every collection run (electronic supplementary material, §S2). Artefacts of those files are reproduced rather than repaired, the consequential one being that the Arabic and Chinese versions state the four cells as penalties but phrase the goal sentence as maximizing a reward, so we treat the five languages as five distinct stimuli rather than five renderings of one.

Each agent also receives one line naming its persona, in the language of the prompt: in English, *You are cooperative.* or *You are selfish.* The two are crossed in all four ordered pairings, ten replicates each. The opponent’s persona is never disclosed (electronic supplementary material, §S2), so a pairing reaches a player only through realised behaviour and not before round 2, and we treat it as a design factor and a regression covariate rather than as information the agent held.

Every game runs ten rounds with simultaneous moves: both agents see the same history, rounds 1 to $r - 1$, and the round is scored once both have replied. Sampling runs at temperature 1.0, under a common-random-numbers scheme that gives a given round of a given replicate the same seed at every payoff scale (electronic supplementary material, §S2.1).

Decisions are elicited as free text and parsed by the protocol’s own matcher. A reply it cannot resolve is regenerated, up to two further attempts, and if all three fail the decision is recorded as OptionA, which under this framing is defection: the fallback can only push a measured cooperation rate down, never up, and it cannot manufacture the cooperative behaviour reported here. Each collection run asserts that its fallback rate is at most 0.02, but the achieved rates are not carried in the released per-game tables, so we cannot report how they vary with the payoff scale (electronic supplementary material, §S2.2).

2.3 Strategy read-out

The read-out reports which of four canonical rules, AllC, AllD, Tit-for-Tat and Win-Stay-Lose-Shift, a ten-round trajectory most *resembles*. It is not an identification of the strategy a model is running, and no claim in this paper treats it as one.

All four rules are memory-one, so exact matching is a lookup with no learning in it, and it returns a *set* rather than a label, because over ten rounds several rules routinely prescribe the same moves. The three provenances, the agent-games matched exactly by one rule (deduced), by several at once (ambiguous) and by none (unmatched), are recorded separately and never merged, and the same matcher yields a continuous quantity needing no labels at all, the number of rounds violating the nearest rule.

Where the rules leave the answer open a single-layer LSTM [61] arbitrates. Trained on the synthetic corpus alone, which contains no language-model output, it reaches 98.5% accuracy in distribution and 93.6% and 81.1% on two execution-noise levels held out entirely (electronic supplementary material, §S6). It ranks only within the set the rule base returns, so it can never overturn a deduction; it decides alone only where no rule fits, and its share of the labels is very unevenly spread across the four rules, which bounds how the composition may be read (§3).

2.4 Statistical analysis

The cooperation rate of an agent-game is the share of its ten decisions that are OptionB; a cell mean averages the agent-games of a cell, and round-level analyses use the binary decision itself. Because the two agents of a dyad play each other and the ten rounds inside an agent-game are strongly autocorrelated, every interval was computed by resampling whole dyads with replacement, 1,500 draws, percentile method [62], and every regression clustered its standard errors on the dyad (electronic supplementary material, §S2.3). Because the range over the ten scales is a maximum minus a minimum, and so positive by construction, its bootstrap interval quantifies the precision of the range and not its distance from zero; the permutation test supplies the test.

Because the null is a theorem, the matching test destroys the association with the payoff scale and leaves the rest of the design as observed: the scale label is shuffled among whole dyads, the unit to which the scale was assigned, and the range of the ten scale means recomputed (electronic supplementary material, §S2.3).

For each model we fitted a logistic regression of the round-level decision on the payoff scale as a ten-level factor, with prompt language, own persona and opponent persona as covariates, $\lambda = 1$ as reference and standard errors clustered on the dyad, and tested the nine scale contrasts jointly by a Wald test (table 1). The variance decomposition is a type-II analysis of variance on the 250 cell means of the five models reported here, with terms for model, language and payoff

scale and their three two-way interactions and no three-way term, so the residual on 144 degrees of freedom is the omitted three-way interaction; the same decomposition on all 300 cells is in the electronic supplementary material, §S4.

Every payoff printed is at most 1 exactly when $\lambda \leq 0.1$, which two of the ten scales satisfy. The persona result contrasts those two scales with the remaining eight, as a difference of differences with an interval from the same dyad bootstrap; where a trend across the whole grid is wanted instead, $\log_{10} \lambda$ enters as a continuous predictor. Analyses were run in Python, and the package versions are listed in the electronic supplementary material, §S9.

3 Results

Across the 240,000 recorded decisions of the whole corpus, the sixth model included, cooperation is 0.5895; the five models reported below carry 200,000 of those decisions.

3.1 Cooperation moves under a transformation that cannot matter

Cooperation depends on the payoff scale in all five models (figure 1b, table 1). The range across the ten scales runs from 0.081 (95% CI 0.062, 0.135) in Claude Haiku 4.5 to 0.636 (95% CI 0.588, 0.686) in Grok 4.20.

The observed range falls in the extreme upper tail of the permuted distribution in every model ($p = 0.0005$ in four of the five, the smallest value 2,000 draws can return, and $p = 0.0025$ in Claude Haiku 4.5), and the per-model logistic regression rejects the joint null on the nine scale contrasts ($\chi^2_9 = 32.3$ to 1227.3 , all $p \leq 2 \times 10^{-4}$), the largest contrast against $\lambda = 1$ ranging from 0.36 to 3.74 in log-odds (table 1).

The shapes are unlike one another. Grok 4.20 cooperates 0.274 at $\lambda = 0.01$, the lowest rate these data contain, and 0.910 at $\lambda = 2$, the highest: the same model, shown the same game, is predominantly a defector at the bottom of the grid and predominantly a cooperator at the top. The others move less, and disagree with it and with each other. GPT-5.4 Nano rises steeply from 0.438 at $\lambda = 0.01$ and saturates near 0.65; Claude Haiku 4.5 dips to a minimum at $\lambda = 0.25$ and recovers; Gemini 3.5 Flash-Lite oscillates within a high band; Qwen3 235B-A22B drifts downward across the whole range.

That disagreement has a direct consequence for reporting. Averaging the five model curves gives a pooled curve whose range is 0.154, while the mean within-model range is 0.249, an attenuation of 1.62 times (figure 1c). The pairwise correlations between model curves average -0.10 and run from -0.72 to $+0.85$, with six of the ten pairs negative. Curves that move in opposite directions cancel when added.

A variance decomposition on the 250 cell means makes the same point in another currency. The model accounts for 45.4% of the variance across cells and the model-by-language interaction for 17.0%. The payoff scale carries 5.8% as a main effect and is significant ($F_{9,144} = 10.64$, $p < 10^{-3}$), but its interaction with the model carries 17.9%, three times as much ($p < 10^{-3}$), so most of what the payoff scale does is specific to the model rather than shared across models. Its interaction with the prompt language carries 3.6% ($p = 0.018$), the weakest test reported here.

That the main effect is significant at all is a fact about one model rather than about language models. It is carried by Grok 4.20, whose departure is large enough to survive being averaged with four others; a panel assembled without that model returns no significant main effect, with

one game, three renderings

a

$\lambda = 0.01$

	opp A	opp B
you A	0.06	0
you B	0.1	0.02

$\lambda = 1$

	opp A	opp B
you A	6	0
you B	10	2

$\lambda = 1000$

	opp A	opp B
you A	6000	0
you B	10000	2000

penalties, to be minimized.
identical preferences, best replies,
equilibria and replicator dynamics.

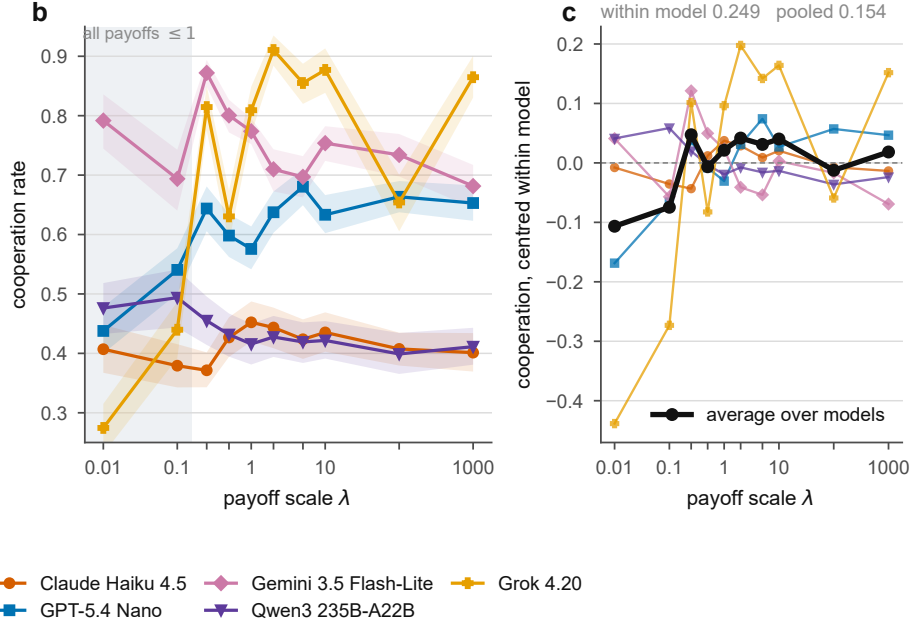


Figure 1: Cooperation against the payoff scale in five language models. (a) The same prisoner’s dilemma at three payoff scales, shown to the agent as penalties to be minimized. (b) Cooperation against the payoff scale, one curve per model; the shaded band marks the two scales at which every payoff printed is at most 1. (c) The curves of (b) centred within model, with their average in black. Each point in (b) and (c) averages 200 dyads, with 95% percentile intervals from 1,500 bootstrap resamples of whole dyads.

every remaining model still moving. A pooled analysis therefore reports a quantity whose significance turns on which models happen to be in the panel, and whose size, 0.154 against a typical within-model 0.249, matches no member of it.

3.2 The size of the scale effect depends on the prompt language

Neither prompt factor outside the game is neutral with respect to the payoff scale. Both the level of cooperation and its sensitivity to the scale differ with the prompt language (figure 2). Level differences are large and model-specific, which is why the model-by-language interaction dominates the decomposition above: Claude Haiku 4.5 cooperates 0.574 in Chinese and 0.234 in Vietnamese, while GPT-5.4 Nano cooperates 0.798 in English and 0.458 in Arabic.

One caveat attaches to the level comparison rather than to the scale result. The Arabic and Chinese templates, reproduced unaltered from the protocol, state the four cells as penalties while phrasing the goal sentence as maximizing a reward (§2), so a level difference between two languages here is a difference between two stimuli and not between two renderings of one. It cannot produce a dependence on the payoff scale, because the wording is identical at every scale, and the ranges below are computed within a language.

The sensitivity differences are the new observation. The range of cooperation across the ten scales itself depends on the language: 0.130 in Vietnamese against 0.314 in French for Claude Haiku 4.5, and 0.162 in Chinese against 0.412 in English for Gemini 3.5 Flash-Lite. Across the 25 model-language pairs it runs from 0.083, in Qwen3 235B-A22B in French, to 0.855, in Grok 4.20 in English. Scale sensitivity is a property of the model and the language together, not of the model alone.

Table 1: Cooperation over the ten payoff scales, by model. Minimum, maximum and range, with a 95% confidence interval on the range from a bootstrap over whole dyads and a permutation test on the payoff-scale label (0.0005 is the floor of 2,000 draws). The last two columns test the nine scale contrasts jointly in a per-model logistic regression of the round-level decision, clustered on the dyad. Each entry pools 200 dyads per scale; ranges are computed before rounding. Full specification in the electronic supplementary material.

Model	cooperation			95% CI	p_{perm}	Wald test	
	min	max	range			χ^2_9	p
Claude Haiku 4.5	0.371	0.452	0.081	[0.062, 0.135]	0.0025	32.3	< 0.001
GPT-5.4 Nano	0.438	0.680	0.243	[0.200, 0.290]	0.0005	164.1	< 0.001
Gemini 3.5 Flash-Lite	0.682	0.872	0.190	[0.165, 0.240]	0.0005	129.0	< 0.001
Qwen3 235B-A22B	0.399	0.494	0.095	[0.068, 0.158]	0.0005	96.9	< 0.001
Grok 4.20	0.274	0.910	0.636	[0.588, 0.686]	0.0005	1227.3	< 0.001

Table 2: Persona effect below and above the sub-unit boundary, by model. Cooperation under a cooperative persona minus cooperation under a selfish one, at the two scales where every printed payoff is at most 1 ($\lambda \leq 0.1$) and at the eight where some payoff exceeds it. Intervals are 95% confidence intervals from the dyad bootstrap. A sign flip needs both regime intervals to exclude zero on opposite sides, so a supra-unit interval straddling zero means the effect is abolished rather than reversed.

Model	persona effect, with 95% CI		shift	95% CI	sign flip
	$\lambda \leq 0.1$	$\lambda \geq 0.25$			
Claude Haiku 4.5	−0.222 [−0.258, −0.186]	+0.207 [+0.189, +0.226]	0.429	[0.389, 0.471]	yes
GPT-5.4 Nano	−0.135 [−0.179, −0.091]	+0.018 [−0.001, +0.037]	0.153	[0.105, 0.200]	no
Gemini 3.5 Flash-Lite	−0.456 [−0.507, −0.406]	+0.224 [+0.207, +0.242]	0.680	[0.625, 0.736]	yes
Qwen3 235B-A22B	−0.911 [−0.931, −0.893]	−0.679 [−0.694, −0.664]	0.233	[0.210, 0.258]	no
Grok 4.20	−0.441 [−0.485, −0.396]	−0.303 [−0.326, −0.279]	0.138	[0.090, 0.185]	no

3.3 The payoff scale gates the persona instruction

The difference between the cooperation rates of the two persona instructions measures how much the persona takes hold. It depends on the payoff scale, and not gradually (figure 3, table 2). The natural boundary is where the printed numbers stop being sub-unit, at $\lambda \leq 0.1$ (§2). The persona effect shifts across it in all five models and every shift excludes zero, from 0.138 (95% CI 0.090, 0.185) in Grok 4.20 to 0.680 (95% CI 0.625, 0.736) in Gemini 3.5 Flash-Lite.

Whether the sign reverses is the stronger claim, and we hold it to a test on each regime rather than to a comparison of two point estimates. Two models of five reverse: in Gemini 3.5 Flash-Lite and Claude Haiku 4.5 the sub-unit interval lies wholly below zero and the supra-unit interval wholly above it, so below the boundary these models cooperate *more* when told they are selfish. In GPT-5.4 Nano the supra-unit interval straddles zero, so its effect is abolished at the boundary rather than reversed, and the other two models keep their sign throughout.

The boundary appears to be one of magnitude rather than of decimal notation. At $\lambda = 0.25$ the payoffs are 0, 0.5, 1.5, 2.5: still decimals, but no longer all below 1, and there the persona effect already has its supra-unit sign and size in both reversing models. The grid does not cross notation with magnitude fully, since every sub-unit scale in it is decimal-printed too.

Qwen3 235B-A22B and Grok 4.20 invert the persona instruction at every scale rather than

Table 3: Rule agreement and rule distance against the payoff scale, by model. Left, the share of agent-games matched exactly by one canonical rule, below and above the sub-unit boundary, with the coefficient of a logistic regression of that indicator on $\log_{10} \lambda$; right, the mean number of rounds violating the nearest canonical rule at the smallest and largest scale, with the same trend by least squares. Both β are per decade of λ and their signs differ between models. Both cluster on the dyad, and neither block uses the learned classifier.

Model	exactly one rule fits (%)				rounds violating the nearest rule			
	$\lambda \leq 0.1$	$\lambda \geq 0.25$	β	p	$\lambda=0.01$	$\lambda=10^3$	β	p
Claude Haiku 4.5	17.9	3.5	-0.545	< 0.001	1.70	2.02	+0.073	< 0.001
GPT-5.4 Nano	9.8	7.6	-0.038	0.441	2.02	2.19	+0.038	0.052
Gemini 3.5 Flash-Lite	32.2	15.6	-0.242	< 0.001	0.48	1.36	+0.194	< 0.001
Qwen3 235B-A22B	72.1	53.6	-0.267	< 0.001	0.27	1.14	+0.202	< 0.001
Grok 4.20	44.6	17.1	-0.301	< 0.001	0.92	0.37	-0.118	< 0.001

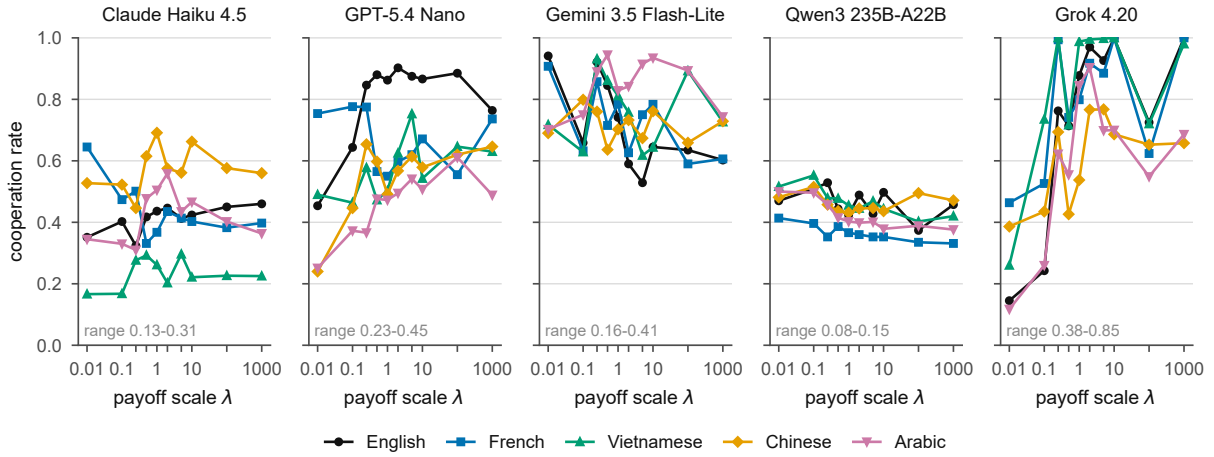


Figure 2: Cooperation against the payoff scale, by model and prompt language. One panel per model, one curve per prompt language; the annotation gives the smallest and largest across-scale range among that model’s five languages. Each point averages 40 dyads.

only below the boundary (table 2), apparently resolving the story’s ambiguity about who one cooperates with, one’s partner or the authorities, opposite to the convention the payoff matrix encodes. The inversion is not our subject, but its magnitude depends on the payoff scale like everything else measured here (electronic supplementary material, §S5).

3.4 The scale also changes what is played

Two models with the same cooperation rate can be playing different games, so we read each agent-game against the four canonical memory-one strategies (§2). What the read-out returns is a resemblance and not an identification, and the labels below should be read that way throughout. Over the whole corpus, all six models included, 23.0% of agent-games are matched exactly by one rule, 17.5% by several at once and 59.6% by none.

Play becomes less rule-governed as the payoffs grow in four models of five, and shows no trend in the fifth (figure 4a, table 3). Pooling the five, the share matched exactly by one canonical rule falls from 38.6% at $\lambda = 0.01$ to 17.3% at $\lambda = 10^3$, and the share matched by none rises from 51.6% to 64.0%. The trend is significant and negative in Claude Haiku 4.5 ($\beta = -0.545$

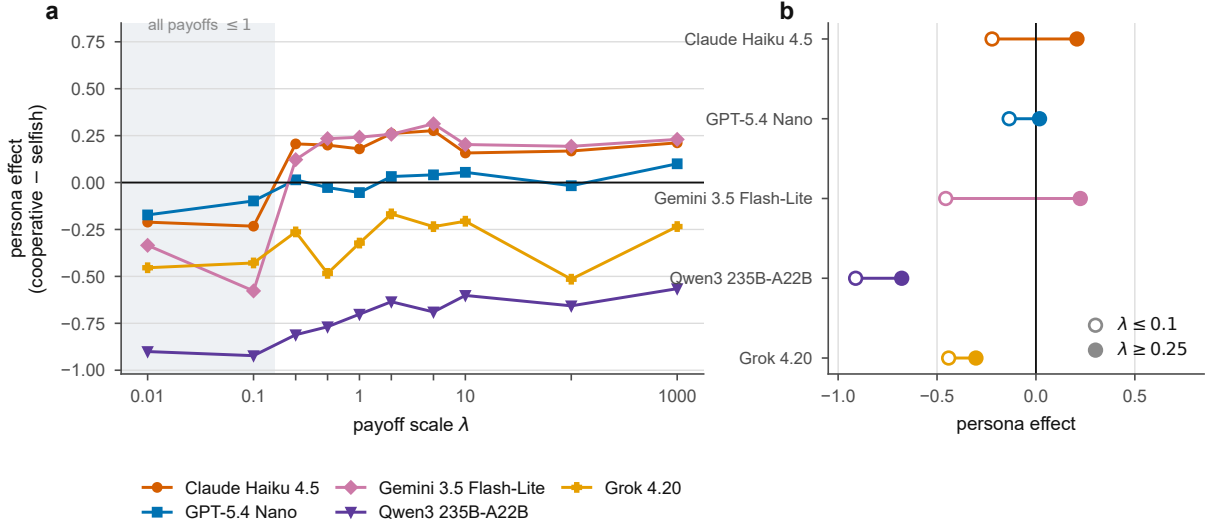


Figure 3: The persona effect against the payoff scale. The persona effect is the cooperation rate under a cooperative persona minus that under a selfish one. (a) The effect against the payoff scale, each point averaging 200 dyads; the shaded band marks the two scales at which every payoff printed is at most 1. (b) The same contrast as two regime means: open markers at $\lambda \leq 0.1$, filled markers at $\lambda \geq 0.25$ (table 2).

per decade of λ), Grok 4.20 (-0.301), Qwen3 235B-A22B (-0.267) and Gemini 3.5 Flash-Lite (-0.242), and absent in GPT-5.4 Nano ($p = 0.44$).

No model in the main text runs the other way on this measure, and one model in the corpus does, so the claim should not be read as uniform. The sixth model, Gemini 3.1 Flash-Lite, which is reported in the electronic supplementary material because its identifier names an unpinned preview endpoint rather than because of anything in its results (§2), carries a significant positive trend ($\beta = +0.125$), its play becoming *more* consistent with a canonical rule as the payoffs grow. It is the only model of the six that does. Over the whole corpus the result is therefore four models moving one way, one showing no trend and one moving the other, and that is the form the evidence supports.

The continuous version needs no labels at all: the number of rounds violating the nearest canonical rule (figure 4b). It rises from 0.48 to 1.36 in Gemini 3.5 Flash-Lite and from 0.27 to 1.14 in Qwen3 235B-A22B between the smallest and largest scale, with significant positive trends there and in Claude Haiku 4.5. Grok 4.20 moves the other way, falling from 0.92 to 0.37 ($\beta = -0.118$, $p < 0.001$): within the main five, three drift away from the canonical rules as the payoffs grow and one drifts towards them.

The composition shifts too, and the models do not shift together (figure 5). Pooled across the five, the AllC share rises from 38.1% to 45.8% of agent-games between the extreme scales and the Tit-for-Tat share from 9.1% to 12.7%, both at the expense of AllD, which falls from 43.6% to 28.3%. That pooled figure conceals the same cancellation: GPT-5.4 Nano moves from 29.8% to 50.7% AllC while its AllD share falls from 46.0% to 17.5%, whereas Gemini 3.5 Flash-Lite moves the other way on AllC, from 73.8% to 51.5%, with Tit-for-Tat roughly doubling from 6.8% to 14.2%.

This last result leans hardest on the learned component, and the provenance is not uniform across the four labels: AllC and AllD are anchored in part by exact matching, whereas 94.6%

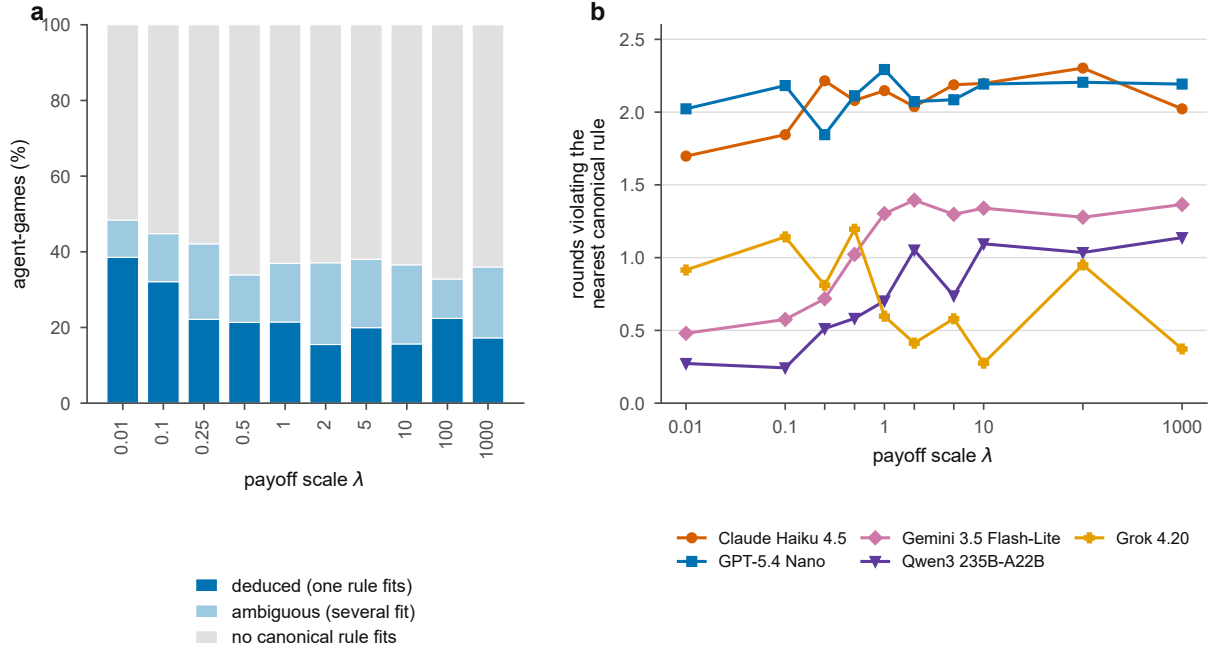


Figure 4: How closely play follows the canonical memory-one rules, against the payoff scale. Both panels are computed by exact matching, with no learned component. (a) Provenance of the strategy read-out, pooled over the five models; the per-model trends, which do not all have the same sign, are in table 3. (b) Mean number of rounds, out of the ten played, violating the nearest canonical rule.

and 97.1% of the Tit-for-Tat and Win-Stay-Lose-Shift labels are attributions made where no canonical rule fits (electronic supplementary material, §S6.2). A movement in the Tit-for-Tat share is therefore a statement about which canonical rule the play most resembles, not a claim that the model is running Tit-for-Tat.

3.5 The effect is present before any strategic interaction

A rescaling could act through the course of play, by changing how a model reads its opponent’s history, or on the reading of the matrix itself. The opening move separates the two: at round 1 nothing but the matrix and the instructions has been seen.

The effect is already present at round 1 (figure 6): in four models of five the range of opening-move cooperation across the ten scales exceeds the range over rounds 2 to 10, by factors of 2.0 in Claude Haiku 4.5, 1.9 in Gemini 3.5 Flash-Lite, 1.3 in GPT-5.4 Nano and 1.2 in Grok 4.20; the exception is Qwen3 235B-A22B. The two windows rest on 400 and 3,600 decisions per cell, so their ranges are not variance-matched and these ratios are descriptive rather than tests.

The reading we take from this is deflationary about dynamics. Whatever the payoff scale does, most of it is already done before the game has begun. That is the timing an initial disposition read off the magnitudes of the matrix would produce. It does not rule out a further effect on how history is weighed, and this design does not separate the two.

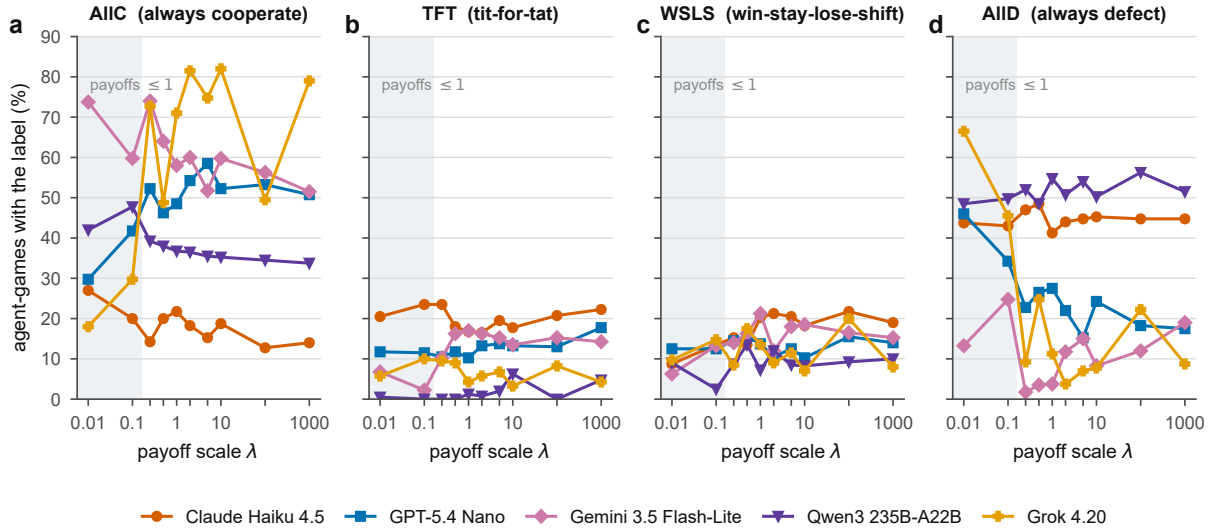


Figure 5: Share of agent-games carrying each canonical label, against the payoff scale. One panel per canonical rule: (a) AllC, (b) Tit-for-Tat, (c) Win-Stay-Lose-Shift, (d) AllD. Each point is the share of that model’s 400 agent-games at that scale carrying the label, and shaded bands mark the two scales at which every payoff printed is at most 1. The AllC and AllD panels rest partly on exact rule matching, whereas the Tit-for-Tat and Win-Stay-Lose-Shift labels are 94.6% and 97.1% attributions made where no canonical rule fits, and should be read as a resemblance rather than as an identification.

4 Discussion

4.1 Payoff-scale invariance is violated

The measurement reported here is unusual in having a null that is a theorem. That property lets a movement in cooperation be read as a departure from an invariance rather than as the difference between two conditions whose expected gap was never specified. The predicted difference between a matrix and the same matrix with larger cells is not small, or contested. It is zero.

All five models depart from it, and none is close to invariant. What makes the departures hard to set aside is not their size but their structure: they are present on the opening move, before any history exists, their size depends on the language the game is posed in, they gate the persona instruction at a boundary of magnitude, and they change what the transcripts look like as sequences. Sampling variability alone is difficult to reconcile with a pattern of that shape. The models appear to be responding to something in the matrix other than the game it encodes, though this design does not identify what.

4.2 Heterogeneity, and why pooled designs miss it

The pooling result deserves emphasis, because it is the mechanism by which an effect of this size could have gone unnoticed. The cause is arithmetic rather than statistical: the shapes disagree, and several pairs are anticorrelated. Heterogeneous effects that cancel are invisible to pooled designs by construction.

In this panel the cancellation is partial rather than total. A main effect of the payoff scale does survive the averaging, and it is significant, but it carries a third of the variance its interaction with the model carries, and the pooled range understates the typical model by a factor of 1.62:

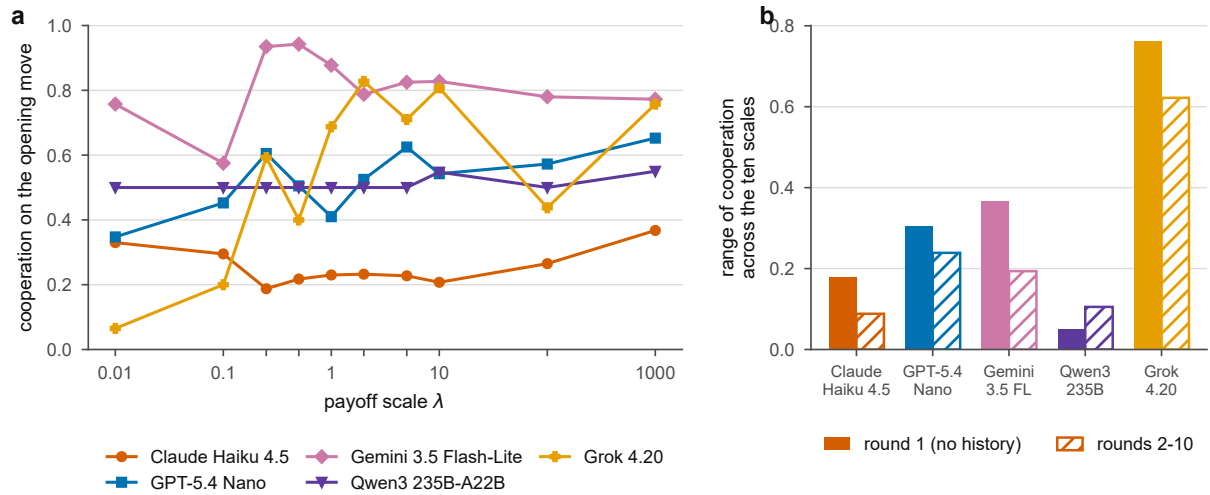


Figure 6: Cooperation on the opening move, against the payoff scale. At round 1 the agent has seen only the matrix and the instructions. (a) Cooperation on the opening move, one curve per model. (b) Range of cooperation across the ten scales, on the opening move (solid) and over rounds 2 to 10 (hatched).

a statement about an average model that nobody deploys.

A conclusion that changes when one model joins or leaves a panel is not a conclusion about language models. The pooled estimate is therefore not a weaker version of the per-model result but a different quantity, and it is not a question of power either: more dyads or more models only tighten the interval around an average of curves that point in different directions. The remedy is a per-model analysis, not a larger sample.

The same logic runs one level deeper: the effect depends on the prompt language as well as on the model, so a per-model number pooled over languages is the same trick played on a smaller stage. The heterogeneity is not a nuisance to be averaged away. It is the finding.

4.3 Implications for behavioural benchmarking of language models

A cooperation rate published for a language model is currently understood as a property of the model. On this evidence it is a joint property of the model, the units the matrix was written in, and the language the game was posed in. The units are the awkward member of that list, because they are the one nobody is asked to record: framings get described, personas get quoted, prompts get reproduced in appendices, while the payoff scale is fixed once, silently, on the authority of a theorem saying the choice is empty. Two laboratories running the same protocol on the same model, with matrices differing by a constant factor, could report cooperation rates differing by as much as 0.636.

We would suggest that a scale-sensitivity measurement accompany a published cooperation rate, in the way an error bar does, and be measured per model, most of all where such rates leave the benchmark and become the parameters of a population model.

4.4 Interpretation and possible mechanisms

Two observations constrain what an explanation must account for. The boundary gating the persona instruction appears to be one of magnitude rather than of notation, a separation only

as strong as the single scale that supports it (§3); and the payoff scale moves how closely play tracks any canonical memory-one rule at all, and shifts the mix of labels with it. The scale is not turning a single dial marked cooperativeness; it is changing the shape of the trajectory.

Several accounts are consistent with both. The printed magnitudes may be read as stakes, which is exactly what a rescaling does not change; numerals of different magnitudes are tokenized differently, so a representation that changes with the digit string could change what follows from it with no notion of stakes involved; training text may associate large quantities with caution; and sub-unit payoffs may fall outside the range in which payoff-like numbers are ordinarily met.

We can order none of these. The design measures departures from an invariance; it does not manipulate tokenization, vary notation at fixed magnitude beyond the one contrast above, or move the framing that supplies the stakes. Nothing here should be read as identifying a mechanism.

There is a constructive reading as well as a cautionary one. A programme is under way to widen the utility function beyond the economic consequences of an action, so that the language describing the action enters it directly, with sentiment analysis proposed as the measurement and the replicator dynamics named as the destination [53]. Our result sits one level below that programme rather than beside it. The evidence for language-based preferences comes from manipulations that could legitimately have mattered, since a label carries connotation, whereas a positive rescaling carries none and is inert by theorem, and the behaviour still moves. A utility that mixes a payoff with a score computed from language is no longer scale-free in the payoff argument unless the two terms are rescaled together, so before such an object is carried into an evolutionary dynamic its invariance class has to be stated. What these data say is that at least one class of agents does not behave as though the payoff argument were scale-free, and that anything built to describe them will have to account for it.

4.5 Limitations and future work

This design does not identify the mechanism inside the model that produces the departures it measures across a broad but finite grid. Six models cannot support claims about model families or about parameter count, and what generalizes here is that the shapes differ rather than what any one of them is. The corpus is one game, one prompt template family and one horizon, so whether the same curves appear in other dilemmas or in longer play is untested.

A sixth model, Gemini 3.1 Flash-Lite, was run on the identical grid and is reported in full in the electronic supplementary material rather than here, because its identifier names an unpinned preview endpoint and what was served under that name during collection cannot be recovered (§2). Two consequences are worth stating rather than leaving to be discovered. First, it is the only model of the six whose strategy trend is positive, so the four-of-five statement in §3 reads, over the whole corpus, as four models of six moving one way, one showing no trend and one moving the other; the second form is the one the corpus supports. Second, including it changes no conclusion here: the all-six pooled range, attenuation, shape correlations and variance decomposition are in the electronic supplementary material, §§8, and every sign, significance and ordering in them agrees with the five-model versions reported above.

The panel dependence of the pooled main effect is a limitation of this study as much as an argument about others. Our own pooled estimate would have been a null had one model been absent from the panel, and it is not a quantity we would defend for its own sake.

One feature of the stimuli bounds the language comparison. Two of the five templates state the payoffs as penalties while phrasing the goal as maximizing a reward, and we reproduced rather than repaired them so that the corpus matches the published protocol. Because that wording is fixed across the ten scales it cannot generate a scale effect, and the within-language sensitivities stand; it does mean that a difference in level between two languages is not attributable to the language alone.

Our reference for calling a departure large is the sampling variability of these data, a bootstrap over dyads and a permutation null, rather than an independent repeat of the whole protocol. A measured test-retest floor, from sweeps re-run under identical conditions, is the obvious next measurement.

The strategy read-out carries a further caveat, which is why its provenance is reported wherever a label appears (§2). A strategy is a decision rule and not a decision, so it is never observed directly, and even where the resident rule is fixed by construction, separating the memory-one rules from observed play has been shown to need a long run of observations [38]. Our trajectories are ten rounds long. The claim that play becomes less rule-governed rests on exact matching and no learned component. The classifier was validated to 20% execution noise, which is two rounds in ten, and in two of the five models play sits about that far from the nearest canonical rule on average, so the read-out is working at the edge of the range it was validated on. The compositional claim uses labels the classifier supplies wherever the rules leave the answer open, is unevenly anchored across the four rules, and is the weaker of the two. Any reading of it as evidence that these models implement Tit-for-Tat would be unsupported by our data.

5 Conclusion

A positive rescaling of a payoff matrix is the rare manipulation whose predicted effect is not merely small but exactly zero. Five frontier language models were played across ten such rescalings, and every one of them moved. The movement is not a uniform bias a calibration could absorb: its shape differs by model and its size by prompt language, so most of it disappears from any analysis that averages models, and what survives the averaging depends on which models happen to be in the average. That is how a departure of this size can sit inside a literature that has not reported it.

What follows is a question about design rather than a verdict on any model. A cooperation rate measured on a language model is reported as though the matrix behind it were an arbitrary and therefore harmless choice. It is arbitrary, and it is not harmless. The same argument extends to every invariance the theory supplies, from shifts of the payoffs to relabelling of the actions, and each is an equally cheap test of whether an agent is playing the game it was shown.

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Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Editors and reviewers can reach all three components of this work now, at [review link]: the analysis pipeline that produces every figure, table and quoted number, with the

tidy tables it writes and the scripts that draw the figures (`Analysis/scaling/`); the transcript corpus of language-model games, held as per-game records of the decisions, payoffs and conditions of every dyad (`Dataset/data_fairgame_frontier_llm/`); and the synthetic corpus that trains the strategy classifier, which contains no language-model output (`Dataset/noise_dataset/`). Raw per-turn request and reply text was not retained. The same three components will be deposited under [licence to be named] in a public archive issuing a citable digital object identifier, [archive DOI], which will be in place at publication. Supplementary material is available online.

Declaration of AI use. Language models are the object of study here: the transcripts analysed were generated by the six models named in §2, through a hosted model-serving platform. Separately, we used a large-language-model coding assistant to help implement the analysis pipeline and to assist with language editing. All analytical choices, statistical specifications and interpretations are the authors’, every number is produced by the archived pipeline and checked against the data by an automated verification script (`Analysis/scaling/s08_verify_paper.py`), and no text, figure or result was generated by an AI system without author verification.

Authors’ contributions. All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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